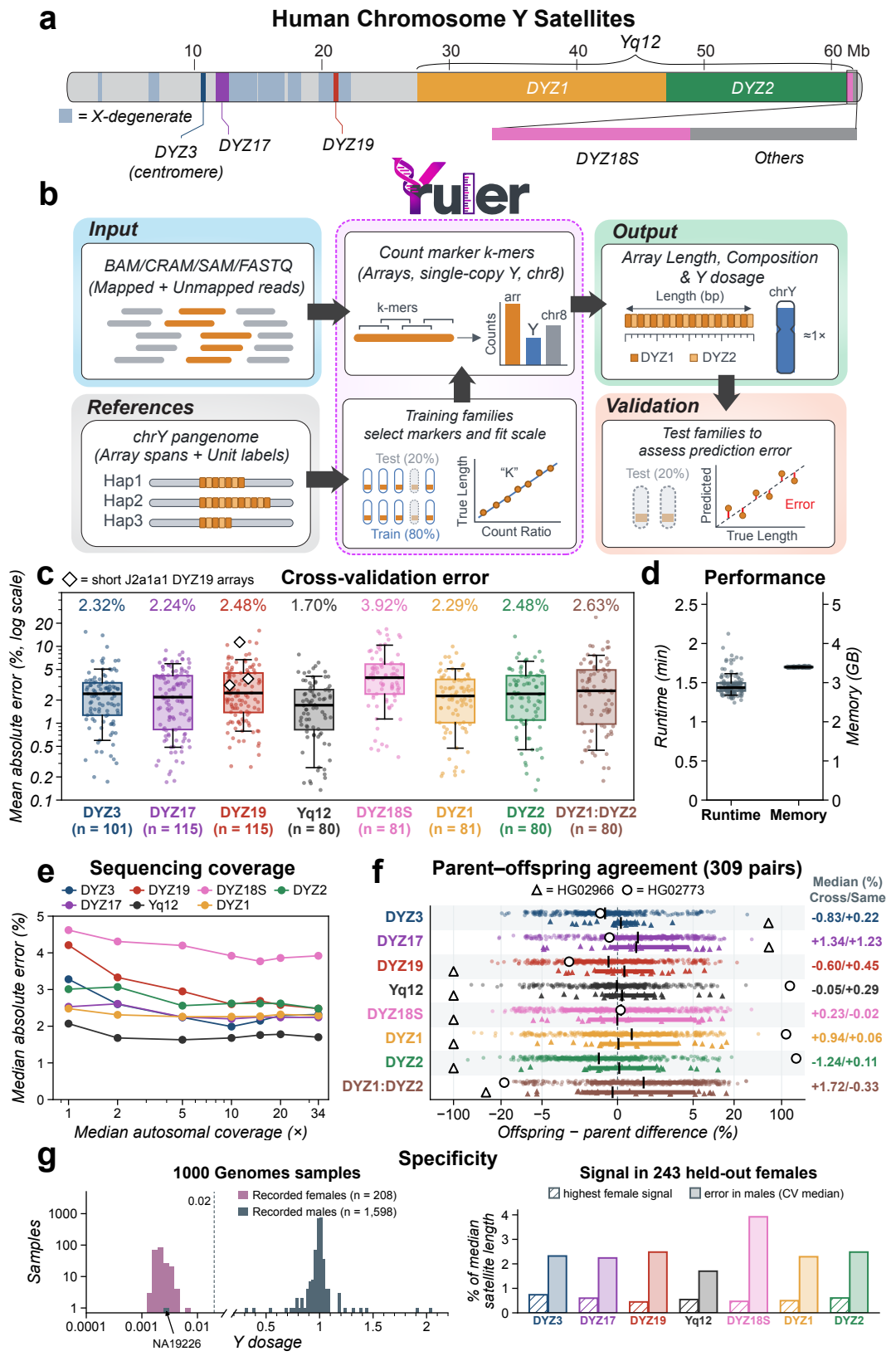

a
Human Chromosome Y Satellites
10
20
30
40
Yq12
50
60 Mb
DYZ1
DYZ2
= X-degenerate
DYZ3
(centromere)
DYZ17
DYZ19
DYZ18S
Others
b
Yuler
Input
BAM/CRAM/SAM/FASTQ
(Mapped + Unmapped reads)
Count marker k-mers
(Arrays, single-copy Y, chr8)
k-mers
Counts
arr
Y
chr8
Output
Array Length, Composition
& Y dosage
Length (bp)
chrY
≈1×
DYZ1
DYZ2
References
chrY pangenome
(Array spans + Unit labels)
Hap1
Hap2
Hap3
Training families
select markers and fit scale
Test (20%)
Train (80%)
True Length
"K"
Count Ratio
Validation
Test families to
assess prediction error
Test (20%)
Predicted
Error
True Length
c
◇ = short J2a1a1 DYZ19 arrays
Cross-validation error
2.32%
2.24%
2.48%
1.70%
3.92%
2.29%
2.48%
2.63%
Mean absolute error (%, log scale)
20
10
5
2
1
0.5
0.2
0.1
DYZ3
(n = 101)
DYZ17
(n = 115)
DYZ19
(n = 115)
Yq12
(n = 80)
DYZ18S
(n = 81)
DYZ1
(n = 81)
DYZ2
(n = 80)
DYZ1:DYZ2
(n = 80)
d
Performance
Runtime (min)
Memory (GB)
2.5
2
1.5
1
0.5
0
5
4
3
2
1
0
Runtime
Memory
e
Sequencing coverage
DYZ3
DYZ19
DYZ18S
DYZ2
DYZ17
Yq12
DYZ1
Median absolute error (%)
5
4
3
2
1
0
1
2
5
10
20
34
Median autosomal coverage (×)
f
Parent–offspring agreement (309 pairs)
△ = HG02966 ○ = HG02773
Median (%)
Cross/Same
DYZ3
-0.83/+0.22
DYZ17
+1.34/+1.23
DYZ19
-0.60/+0.45
Yq12
-0.05/+0.29
DYZ18S
+0.23/-0.02
DYZ1
+0.94/+0.06
DYZ2
-1.24/+0.11
DYZ1:DYZ2
+1.72/-0.33
−100
−20
−5
0
5
20
100
Offspring − parent difference (%)
g
Specificity
1000 Genomes samples
Recorded females (n = 208)
Recorded males (n = 1,598)
0.02
Samples
1000
100
10
1
0.0001
0.001
0.01
0.5
1
1.5
2
NA19226
Y dosage
Signal in 243 held-out females
highest female signal
error in males (CV median)
% of median satellite length
4
3
2
1
0
DYZ3
DYZ17
DYZ19
Yq12
DYZ18S
DYZ1
DYZ2